

Technical Evaluation Report: Ingestion & Inference Architecture

Executive Summary: This document details the infrastructure specifications, model evaluation matrix, and quarterly performance metrics for the unified multimodal ingestion pipeline. The system utilizes distributed GPU nodes with real-time vector indexing.

Figure 1: Multimodal Ingestion Pipeline Flow

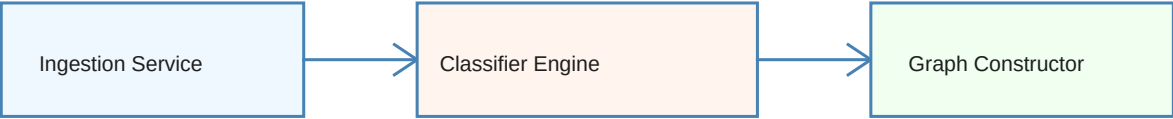


Table 1.1: Serving Node Hardware & Model Specifications (Key-Value)

System Parameter	Configured Value
Target Model Architecture:	Llama-3.3-70B-Instruct
Context Window Size:	128,000 tokens
Quantization Format:	AWQ 4-bit
Tensor Parallelism:	4x NVIDIA H100
KV Cache Allocation:	32 GB PagedAttention

Comparative Analysis: To determine the optimal inference engine, extensive load tests were conducted across vLLM, TensorRT-LLM, and TGI under simulated peak traffic of 10,000 concurrent user sessions.

Table 1.2: Inference Engine Benchmark Matrix (Comparison)

Serving Metric	vLLM 0.6	TensorRT-LLM 0.12	HuggingFace TGI 2.0
TTFT (Time to First Token)	18.2 ms	14.5 ms	22.4 ms
Inter-token Latency (ITL)	8.4 ms	6.2 ms	9.8 ms
Throughput (tokens/sec)	4,850 tok/s	6,120 tok/s	3,950 tok/s
PagedAttention Support	Native	Optimized Kernel	Standard Paged

Temporal Trend Analysis: Production telemetries gathered over the preceding four fiscal quarters exhibit consistent token volume expansion alongside decreasing p99 latencies following kernel optimization deployments.

Table 1.3: Historical Ingestion Throughput & Latency (Time-Series)

Production Metric	Q1 2024	Q2 2024	Q3 2024	Q4 2024
Total Processed Tokens (B)	142.5 B	210.8 B	340.2 B	512.0 B
Average P99 Latency (ms)	84.2 ms	72.5 ms	54.1 ms	42.0 ms
Infrastructure Cost (\$/M Tok)	\$0.45	\$0.38	\$0.29	\$0.22
Service Availability Uptime	99.92%	99.96%	99.98%	99.99%